LABORATORY · Lab · 26 hours

Experiment 9: Design and Implementation of an RC Lag Compensator



Control Systems Laboratory (22EEG53)

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Dr. Mitesh Astik · Dept. of Electrical and Electronics Engineering
NMAM Institute of Technology, Nitte · 2025-26

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Learning objectives

- Design an RC lag network for a given β and T
- Implement it on a bread board and measure the frequency response
- Verify using MATLAB Bode plot
- Determine the maximum phase lag

Syllabus context

Laboratory component of 22EEG53 Control Systems (0:0:2 hours per week). CIE: record and observation 25, viva-voce 5, lab test 20 marks.

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Aim and apparatus

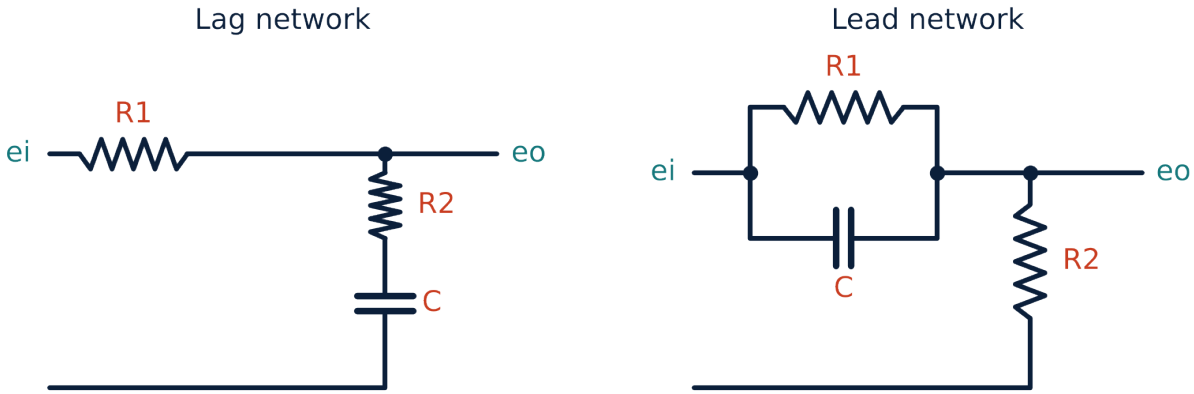
Aim: to design and implement an RC lag compensator and verify the result using MATLAB.

Apparatus: resistors, capacitor, function generator, CRO/DSO, bread board, MATLAB.

Design: β = 10, T = 10 ms.

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RC lag network



Left: lag network

- Gc(s) = (1 + sR2C)/[1 + s(R1 + R2)C].
- T = R2C; β = (R1 + R2)/R2.
- Pole at 1/(βT), zero at 1/T.
- High-frequency gain = 1/β.

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Design calculations

Problem Design a lag network with $\beta = 10$ and $T = 10$ ms. Choose $C = 1\ \mu F$.

- $T = R2C \Rightarrow R2 = 10\ \text{ms}/1\ \mu\text{F} = 10\ \text{k}\Omega$
- $\beta = (R1 + R2)/R2 = 10 \Rightarrow R1 = 9\ R2 = 90\ \text{k}\Omega$
- $G_c = (1 + 0.01s)/(1 + 0.1s)$; corners at 10 and 100 rad/s
- $\varphi_m = -\sin^{-1}[(\beta - 1)/(\beta + 1)] = -54.9°$ at $\omega_m = 1/(T\sqrt{\beta}) = 31.6$ rad/s

Answer: R1 = 90 kΩ, R2 = 10 kΩ, C = 1 µF

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Procedure

1

Build

Assemble the network on the bread board.

2

Input

Apply a sine wave of 1 V peak from the function generator.

3

Measure

For 1–1000 rad/s measure output amplitude and phase (CRO).

4

Plot and verify

Plot gain and phase; compare with MATLAB bode.

Gain = Vout/Vin (dB); phase = −360° × f × Δt.

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MATLAB verification

```
R1 = 90e3; R2 = 10e3; C = 1e-6;
T = R2*C; beta = (R1+R2)/R2;
Gc = tf([T 1], [beta*T 1])
bode(Gc), grid on
[mag, ph, w] = bode(Gc);
[phmin, i] = min(squeeze(ph));
fprintf('Max lag = %.1f deg at %.1f rad/s\n', phmin, w(i))
```

LAB

Observation table

f (Hz)	ω (rad/s)	Vo (V)	Gain (dB)	Phase (deg)
1	6.3			
5	31.4			
20	125.7			
100	628			

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Key takeaways

- Lag: $G_c = (1 + sT)/(1 + s\beta T)$, $\beta = (R1 + R2)/R2$, $T = R2C$
- High-frequency attenuation $20 \log \beta$ dB (20 dB for $\beta = 10$)
- Maximum lag $-54.9°$ at 31.6 rad/s
- Measured response agrees with MATLAB

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Practice questions

Q1. Design a lag network with $\beta = 5$, $T = 20$ ms.

Q2. Find the maximum phase lag of your design.

Q3. What is the high-frequency gain of the network?

Q4. Where is the compensator pole relative to the zero?